Little big lies:
How little data tampering is needed to manufacture statistically significant results

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Abstract

Fabricating a statistically significant result does not require fabricating a dataset. I simulate a dishonest analyst who starts from data generated under the true null hypothesis and alters, deletes, or invents one data point at a time, always the point that most helps, re-running the test after every change until $p < .05$. I formalize this as a family of iterative greedy tampering strategies (dropping cases, switching condition labels, re-pairing values, duplicating cases, fabricating cases, and replacing cases) across three common designs: the two-group mean difference (Student's t-test), the Pearson correlation, and the 2x2 association (chi-square test). Each strategy runs until significance is reached or further tampering becomes structurally impossible, and the estimand is the effort required: the number of tampered data points, both absolutely and as a proportion of the sample. Two findings recur. First, effort is small. Median tamper counts correspond to a few per cent of the sample at realistic sample sizes. The proportion of the sample that must be tampered does not grow with sample size, so large studies are no safer proportionally than small ones. Second, the strategies that are likely hardest to detect, because they preserve every univariate distribution (condition switching; value re-pairing), are among the most effective. I conclude that fabrication and falsification of results typically requires a worryingly small amount of actual data tampering.

In November 2019, the pivotal trial of a treatment for ANCA-associated vasculitis was unblinded and its primary analysis run. The result was not significant ($p = .1025$). According to a subsequent letter from the US Food and Drug Administration, unblinded personnel employed by the sponsor then selected nine of the trial's 331 patients for re-adjudication of their remission status, having first confirmed that reclassifying five of them would move the study across the significance threshold. The five were reclassified, the database was re-locked, and the re-run analysis returned $p = .0132$. That analysis was the only one submitted in support of approval, which was granted in 2021. In April 2026 the FDA proposed to withdraw it, concluding that the integrity of the trial had been compromised; the sponsor contests this and a hearing is pending (U.S. Food and Drug

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Administration, Center for Drug Evaluation and Research, 2026). Whatever its resolution, one feature of the case is worth dwelling on. Five records out of 331, just 1.5% of the sample, separated a failed trial from a successful one.

How often does this happen? Meta-analysis of the survey literature puts self-admitted research misconduct at 2.9% of researchers (95% CI [2.1, 3.8]), comprising 1.9% for fabrication and 3.3% for falsification, while 15.5% report having witnessed it in others (Xie et al., 2021; see also Fanelli, 2009). When the National Survey on Research Integrity used a randomized response design to give respondents provable deniability, admitted fabrication rose to 4.3% and falsification to 4.2% (Gopalakrishna et al., 2022). From the other side of the desk, 24% of consulting biostatisticians in a US national survey had been asked, at least once in the preceding five years, to remove or alter data records to better support a research hypothesis (Wang et al., 2018). These are not the rates of a vanishing phenomenon, and each is likely a lower bound of the true prevalence.

Prevalence, though, is only half the question. The other half is how 'effective' such data manipulations actually are. Here, the literature is largely silent. Chambers (2017) set out a "dirty dozen" of ways to guarantee significant results even under the null hypothesis, the second of which is switching participants between conditions. Brown (2020) posted a short simulation of the efficacy of this in a blog post. Both convey the same intuition that a handful of unjustified changes to the data is enough to produce statistically significant results in many cases, but neither is a systematic assessment. The field has no systematic account of how much tampering a given strategy requires, whether that requirement grows with sample size, how the strategies compare with one another, or how large an effect they leave behind.

For the ambiguous end of the continuum of merely 'questionable' research practices, that gap has been filled. Stefan & Schönbrodt (2023) compiled 12 *p*-hacking strategies from the literature, identified the factors controlling each one's severity, and simulated the resulting increase in false-positive rates. Their scope, as they state it, is the "grey area between good practice and outright scientific misconduct". They stop deliberately at its far edge: discussing misreported test statistics, they note that at some point one must ask whether a practice is still questionable or has become outright fraud. As such, the literature says comparably little about the impact of indefensible research practices on false-positive rates. Of course, given a willingness to engage in an arbitrary amount of data falsification or fabrication, by definition one can obtain statistically significant results. Here, I ask a narrower question: how much data tampering, of what form, is typically needed to acquire significant results.

Quantifying the indefensible practices is worth doing because of how researchers appear to decide what to do. Entradas et al. (2026) find that perceived seriousness is by far the strongest predictor of whether a researcher engages in a questionable practice, ahead of every career and demographic factor they measure. Equally, in John et al. (2012), deciding whether to exclude data after seeing its effect on the results was admitted by 38–43% of psychologists and rated 1.61 out of 2 for defensibility, while falsifying data was admitted by 0.6–1.7% and rated 0.16. These results are consistent with the idea of perceived norms shaping research behaviors. That is, if researchers are more likely to engage in practices they understand to be seriously problematic and are less likely to engage in ones that are seen as such. However, the boundary between these plausibly varies between individuals, research communities, and over time, creating what is effectively an Overton Window of questionable versus unacceptable research practices. Many research practices that were regarded as acceptable prior to the invention of the term *p*-hacking are now considered in many research communities to be substantially less acceptable. If perceptions of how problematic a given set of research behaviors are influence the prevalence of those behaviours, then establishing how relatively bad a given set of practices actually are, how little effort they take, and how much bias they inject, likely represents important information to dissuading their practice.

This perspective is an explicitly social-epistemological one, which understands researcher behaviors as a set of evolving community norms rather than moral acts. Equally, this perspective is concerned only with the undesirable statistical consequences of certain researcher behaviors, but is agnostic to researchers' intent when taking them. With that being said, at a human level, it is easier to imagine a researcher who deletes a small number of data points to 'reveal' an effect they strongly believe is there, lying just under the surface of the data, when it is not (e.g., cases of 'qualitative exclusion strategies': Schiller et al. (2018)) than to imagine scientists simply making up datasets wholesale [although the latter does occur; Verfaellie & McGwin (2011)], despite both strategies producing false-positive results. Regardless of whether these two scenarios can be equated in their intent, it is an open question whether they can be equated in their outcomes, as it is generally unclear how relatively robust or fragile most results are to modifications of only a small number of data points (although see Spiess et al., 2024).

To this end, I invert the design used in previous simulations to understand the impact of *p*-hacking. Where Stefan & Schönbrodt (2023) fix a budget of manipulations and measure the resulting false-positive rate, I fix the goal — statistical significance — and measure the number of tampered data points required to reach it. I take this to be the more ecologically valid framing for deliberate tampering: an analyst who has decided to manufacture a result does not set a budget in advance and stop when it is exhausted, but continues until the *p*-value crosses. It also yields effort directly, in units of edited data points.

In this article I do three things. First, I formalize data tampering as a family of iterative greedy algorithms (dropping, condition switching, re-pairing, duplication, fabrication, and replacement) applied to three of the most common elementary designs: the two-group mean difference (Student's *t*-test), the Pearson correlation, and the 2×2 association (chi-square test). Second, I estimate the effort each strategy requires, with the full design specified under the ADEMP framework (Morris et al., 2019; Siepe et al., 2024) and every estimate accompanied by a Monte Carlo standard error. Because the simulations run without any tampering budget, the complete distribution of required effort is observed, and the proportion of datasets exceeding any given budget is recovered afterwards from the same runs. Third, I show that the strategies hardest to detect are among the cheapest, and that the manufactured effect sizes cluster just above the critical effect size — large enough to publish, too small to look suspicious.

1. Methods

The design is specified under the ADEMP framework (Aims, Data-generating processes, Estimands, Methods, Performance measures) (Morris et al., 2019), as operationalized for psychology by Siepe et al. (2024). The three research compendia containing all simulation code, per-strategy results, and extended rationale are available in the project repository (see *Data and code availability*); this section compresses them.

1.1. Aims

The study is descriptive: there is no closed-form benchmark for the minimum number of tampered points needed to manufacture significance, because that number is a property of a tampering algorithm applied to random data, not an estimator of a population parameter. For each strategy and sample size I aim to estimate (1) how many tampered data points are required to first reach $p < .05$, absolutely and as a proportion of the sample, and how this scales with sample size; (2) the proportion of datasets requiring more than a given share of the sample (post hoc budgets of 10% and 25%), and the proportion for which significance is never attained at all; (3) the effect-size bias of the final tampered dataset relative to the true effect of zero; and (4) how the strategies compare on these measures.

1.2. Data-generating processes

All data are generated under the true null, so every significant result is a false positive produced entirely by the tampering.

- Mean difference. Two independent groups of `n_per_condition` observations each, $Y_{ij} \sim \mathcal{N}(\mu_j, \sigma^2)$ with $\mu_{\text{control}} = \mu_{\text{intervention}} = 0$ and $\sigma = 1$.
- Correlation. A bivariate Normal sample of n observations with $\rho = 0$ and unit variances.
- 2×2 association. Two groups (exposed, not exposed) of `n_per_condition` observations with a binary outcome drawn at the same Bernoulli prevalence in both groups; prevalence is varied (.20 vs .05) to contrast a common with a rare outcome.

Per-group sample sizes are 10 and 25 to 200 in steps of 25 (total N = 20–400; the correlation setting uses the same total Ns). Fully crossing strategy with sample size (and prevalence) gives 45 conditions in each continuous setting and 54 in the binary setting, each replicated 10,000 times.

1.3. Estimands

The primary estimand is `tampers_needed`: the number of tampering operations applied before the test first returns $p < .05$. The secondary estimand is the effect size of the final tampered dataset — Hedges' g in the mean-difference setting and Pearson's r in the correlation setting — which, because the true effect is zero, *is* the bias the tampering introduces; it is reported signed, since the sign is the direction of the manufactured effect. (No standardized effect-size bias is computed in the binary setting.)

1.4. Methods (tampering strategies)

Every strategy is an iterative greedy procedure: generate a null dataset, test it, and while non-significant, apply one tampering operation targeting the data point most opposed to the desired result, then re-test. Table 1 defines the operations. All strategies push toward a positive effect by convention; tests are two-sided at $\alpha = .05$ (equal-variance t-test; Pearson correlation test; chi-square test without continuity correction).

Table 1: The tampering strategies and their operationalization in each setting. Each operation counts as one tampered data point. For binary data, duplication, fabrication, and replacement collapse onto the same atomic operation (adding a favourable cell) and are treated as a single strategy.

Strategy	Mean difference	Correlation	2×2 association
Dropping	Delete the most hostile case (lowest in intervention or highest in control)	Delete the most discordant case (most negative centred cross-product)	Delete a 1 from not-exposed or a 0 from exposed
Label switching	Move the most hostile case's *condition label* to the other group, alternating direction to keep groups balanced	*Re-pairing:* swap the y-values of two cases so the most discordant case receives the y matching its x-rank	Move a not-exposed 1 to exposed or an exposed 0 to not-exposed, alternating direction
Duplication	Duplicate a case from each group's most favourable 10%	Duplicate one of the most concordant cases	(coincides with fabrication)
Fabrication	Add invented cases: control $\sim \mathcal{N}(0, 0.2)$, intervention $\sim \mathcal{N}(5, 0.2)$	Add a high-leverage concordant pair at $\pm(3, 3)$ with small scatter	Add a 1 to exposed or a 0 to not-exposed
Replacement	Replace the most hostile case with an invented one (whichever of the two candidate replacements most lowers p)	Replace the most discordant case's y with a concordant high-leverage value	(coincides with fabrication)

The label-switching column deserves emphasis because of its forensic properties: condition switching leaves every univariate distribution untouched, and re-pairing preserves both marginal distributions *exactly* (the set of y-values is only permuted), so neither is detectable by any univariate check. Swapping *entire* conditions, as opposed to individual cases, is handled analytically rather than simulated: it merely doubles the directional false-positive rate to α, adding $\frac{1}{2}\alpha$.

No tampering budget is imposed. Each strategy runs until significance is reached or a structural limit is hit — the data are exhausted under dropping, no improving re-pairing exists, a donor cell empties in the 2×2 — with a loop guard at 10× the sample size as a termination backstop for pathological greedy stalls (e.g., a one-step greedy cannot cross the two-sided test's $p \approx 1$ valley when the initial dataset happens to show a strong effect in the *opposite* direction; such iterations, well under 1% of runs, are recorded as never attaining significance). Because the full unbudgeted counts are recorded, the proportion of datasets exceeding *any* tampering budget is computable after the fact; I report budgets of 10% and 25% of the sample.

1.5. Performance measures and Monte Carlo uncertainty

An iteration that never attains significance has no finite tamper count — it is right-censored. All percentile-based summaries (median, 95% prediction interval) are therefore computed over all iterations with the never-significant ones ranked above every finite count; a percentile falling among them is reported as *never* rather than being silently replaced by the survivors' percentile, which would answer a different question ("given that tampering worked, how many edits?") and bias the effort downward. The mean cannot be computed this way at all (the censored counts are unbounded above), so means are reported only as conditional-on-success skew diagnostics. The 95% prediction interval is the 2.5th–97.5th percentile of the tamper counts across iterations: the range within which a single new study's required effort would fall 95% of the time.

Every reported estimate carries a Monte Carlo standard error: binomial for proportions, closed-form for means, and nonparametric bootstrap for medians (Morris et al., 2019; Siepe et al., 2024). In tables, decimal places are set from the MCSE (two significant figures, with the estimate rounded to match) rather than fixed per table. Simulations were parallelized with reproducible L'Ecuyer–CMRG streams, so results are identical regardless of worker count.

2. Results

> **Draft status.** This draft was rendered from 16 of the 16 simulation cells (16 from the optimised implementation). The largest tamper count observed anywhere in these runs is 885.0% of the sample, which is **inconsistent** with the superseded design that capped tampering at 50%; treat every number below as provisional until the definitive run is in place.

2.1. Effort is small and proportionally flat in sample size

Across both continuous settings, the absolute number of tampered data points needed to manufacture significance grows with sample size, but the *proportion* of the sample does not (Figure 1, Figure 2): at the largest sample size examined ($N = 400$), the median proportion tampered ranges from 1.0% to 6.0% across strategies. Manufacturing a false positive is no harder, proportionally, in a large study than in a small one. Framed as a budget: at $N = 400$, the proportion of datasets that would require more than 10% of the sample to be tampered is at most 21.4% for the least efficient continuous-setting strategy, and for the cheapest it is 0.0%. Only 21.4% of datasets resist the least effective strategy altogether at that sample size.

The 2×2 setting is the partial exception that illustrates the mechanism (Figure 3). With a rare outcome (prevalence .05) and small samples there is simply not enough signal-bearing data to tamper with: at $N = 20$, 60% of datasets cannot be pushed to significance by dropping at all — the strategy exhausts the table's hostile cells first. Where tampering *is* structurally possible, rare outcomes are cheap to tip, because each fabricated or relabelled case moves the test statistic disproportionately.

2.2. The stealthiest strategies are among the cheapest

Strategies in every figure are ordered by their median proportion tampered, computed from the data at render time. The forensically critical result is where the label-switching family sits in that ordering: condition switching (median 5 switched labels at $N = 100$) and re-pairing (median 8 re-paired values at $N = 100$) reach significance with effort comparable to or lower than the visibly destructive strategies, while leaving every univariate distribution intact — re-pairing preserves both marginals *exactly*. No check of either variable in isolation can detect them; only the joint structure is tampered. Efficiency and detectability are

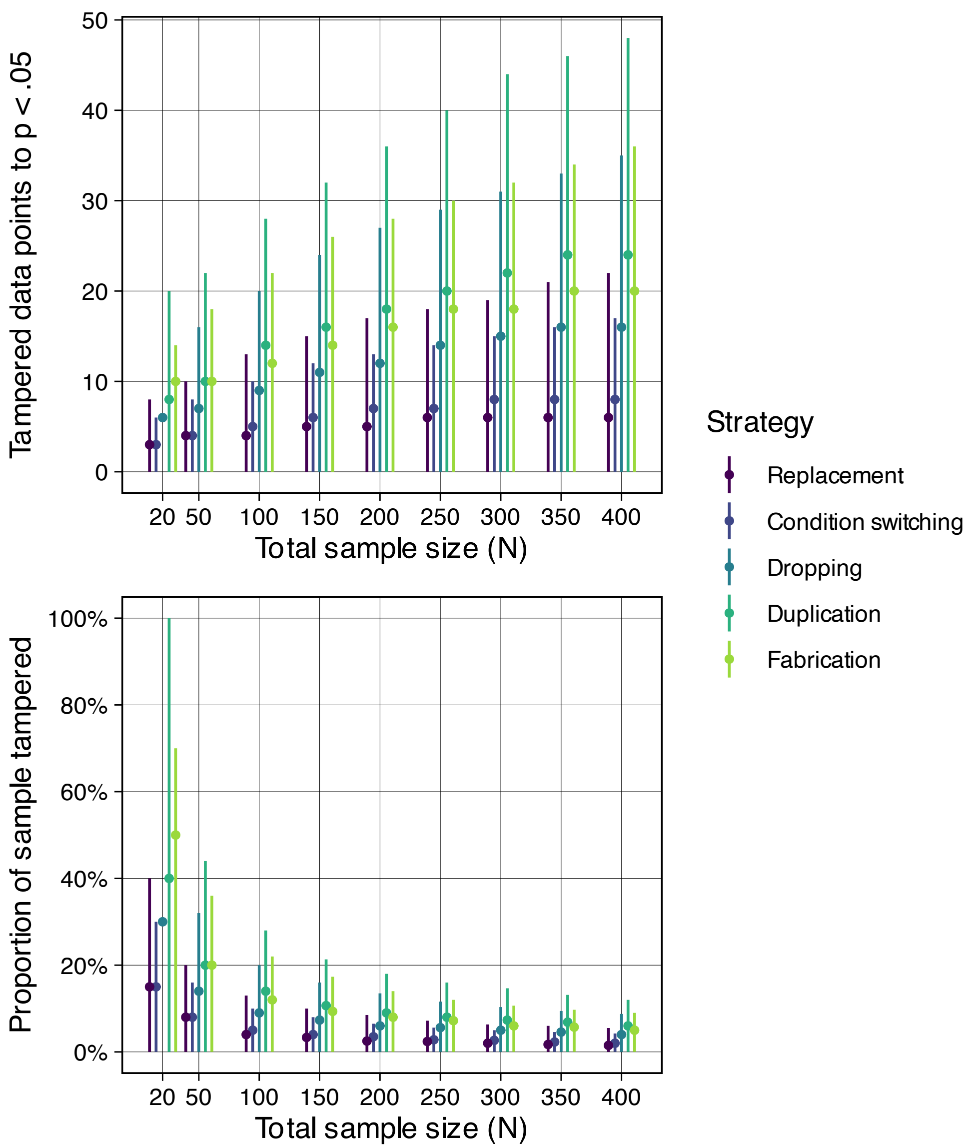

Figure 1: Mean-difference (t-test) setting: the number of tampered data points needed to first reach p < .05 (top) and the same effort as a proportion of the sample (bottom). Points are censoring-aware medians and bars are 95% prediction intervals (2.5th–97.5th percentile across iterations); a bar is omitted where its upper percentile falls among datasets that never attained significance. Strategies are ordered by median proportion tampered. K = 10,000 iterations per condition.

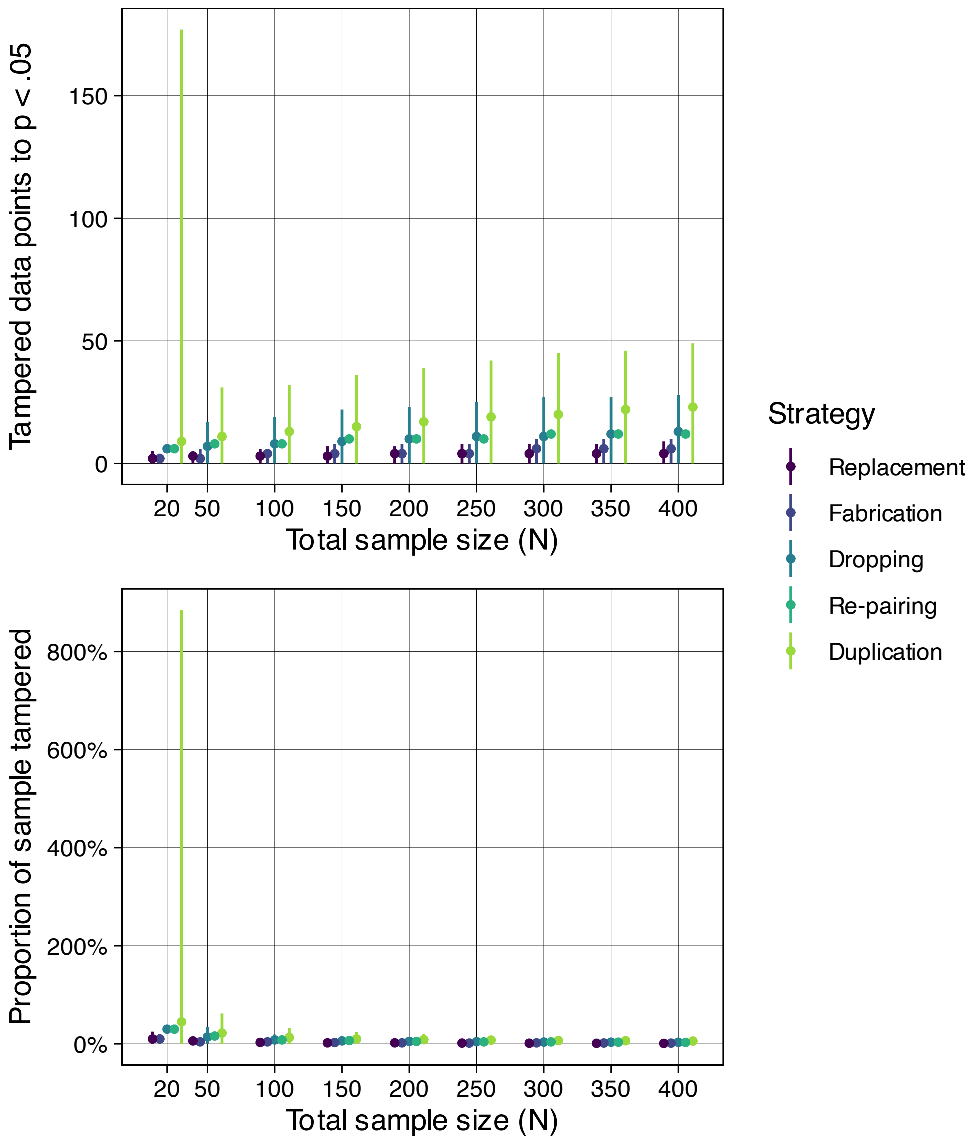

Figure 2: Correlation setting: tampered data points needed to first reach p < .05 (top) and as a proportion of the sample (bottom). Points are censoring-aware medians, bars 95% prediction intervals; a bar is omitted where its upper percentile is never attained. K = 10,000 iterations per condition.

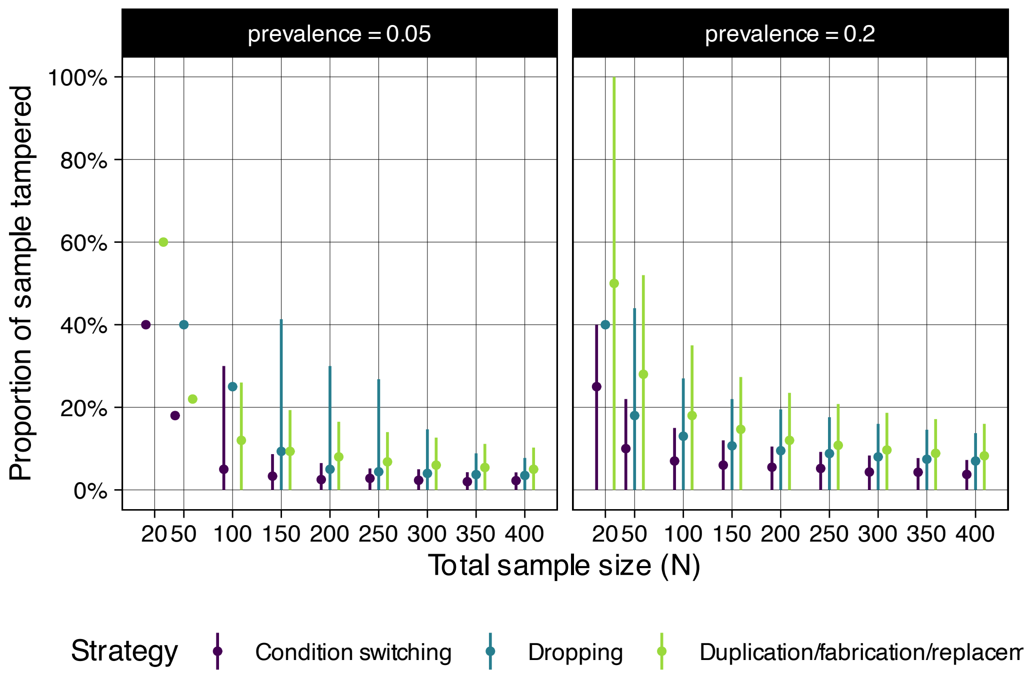

Figure 3: 2×2 association (chi-square) setting, by outcome prevalence: tampered cells needed to first reach $p < .05$ as a proportion of the sample. Points are censoring-aware medians, bars 95% prediction intervals; a bar is omitted where its upper percentile is never attained. At prevalence .05 and small N, a majority of datasets cannot be pushed to significance at all by dropping (the median itself is never attained and no point is drawn). $K = 10{,}000$ iterations per condition.

thus not in tension for the would-be fraudster: the cheap strategies include the invisible ones.

Table 2: Tampering effort at a representative sample size (total N = 100), by setting and strategy. Values in parentheses are <U+00B1>1 Monte Carlo standard error. Never is the proportion of datasets for which significance was never attained, tampering having run to a structural limit. The two budget columns are post hoc tampering budgets: the proportion of datasets needing more than that share of the sample tampered, never-significant datasets exceeding every budget by definition. The median is computed over all iterations with never-significant ones ranked above every finite count, and is also given as a proportion of the sample (Prop.); prediction intervals are shown in the figures. D/f/r denotes the duplication, fabrication and replacement strategies, which coincide for binary data.

Setting	Strategy	Never	> 10%	> 25%	Median	Prop.
Mean difference	Condition switching	0.000 (0.000)	0.020 (0.001)	0.000 (0.000)	5.0 (0.0)	0.050 (0.000)
	Dropping	0.000 (0.000)	0.414 (0.005)	0.001 (0.000)	9.0 (0.0)	0.090 (0.000)
	Duplication	0.000 (0.000)	0.659 (0.005)	0.081 (0.003)	14.0 (0.0)	0.140 (0.000)
	Fabrication	0.000 (0.000)	0.625 (0.005)	0.001 (0.000)	12.0 (0.6)	0.120 (0.006)
	Replacement	0.000 (0.000)	0.062 (0.002)	0.000 (0.000)	4.0 (0.0)	0.040 (0.000)
Correlation	Dropping	0.000 (0.000)	0.345 (0.005)	0.001 (0.000)	8.0 (0.0)	0.080 (0.000)
	Duplication	0.000 (0.000)	0.638 (0.005)	0.102 (0.003)	13.0 (0.2)	0.130 (0.002)
	Fabrication	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	4.0 (0.0)	0.040 (0.000)
	Re-pairing	0.233 (0.004)	0.354 (0.005)	0.233 (0.004)	8.0 (0.0)	0.080 (0.000)
	Replacement	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	3.0 (0.0)	0.030 (0.000)
2<U+00D7>2, prev. 0.05	Condition switching	0.006 (0.001)	0.140 (0.003)	0.034 (0.002)	5.0 (0.0)	0.050 (0.000)
	Dropping	0.057 (0.002)	0.735 (0.004)	0.445 (0.005)	25.0 (0.5)	0.250 (0.005)
	D/f/r	0.006 (0.001)	0.617 (0.005)	0.032 (0.002)	12.0 (0.5)	0.120 (0.005)
2<U+00D7>2, prev. 0.2	Condition switching	0.000 (0.000)	0.263 (0.004)	0.000 (0.000)	7.0 (0.0)	0.070 (0.000)
	Dropping	0.000 (0.000)	0.629 (0.005)	0.040 (0.002)	13.0 (0.0)	0.130 (0.000)
	D/f/r	0.000 (0.000)	0.788 (0.004)	0.224 (0.004)	18.0 (0.2)	0.180 (0.002)

2.3. The manufactured effect is just large enough

Because the true effect is zero, the signed effect size of each final dataset is itself the bias the tampering introduced. Figure 4 shows that the median manufactured effect tracks the critical effect size for the sample size at hand: the greedy procedure stops the moment significance is crossed, so it produces effects just large enough to clear the threshold and no larger. This has an uncomfortable implication for detection: manufactured effects of this kind are, by construction, of exactly the magnitude that a reader would consider unremarkable for the sample size at hand — they look like ordinary, modestly powered findings, and they shrink as N grows just as a real literature's small-study effects are expected to.

The effect-size distribution is a mixture, and its components are worth separating because they explain the negative values visible in Figure 4. Every strategy pushes toward a *positive* effect, so a negative final effect size can arise in only two ways. First, 5.4% of datasets are significant under the null before any tampering — the Type I error rate, as it should be — and those split roughly evenly between the two directions, putting about half that mass below zero. Second, in 0.20% of the 851,641 iterations that did tamper, the greedy procedure stalled before it had turned the effect positive; these are concentrated in re-pairing, whose accept-only-if-r-increases rule can leave it stuck, and **none of them reached significance** (405 of 1,720). No manufactured false positive is negative.

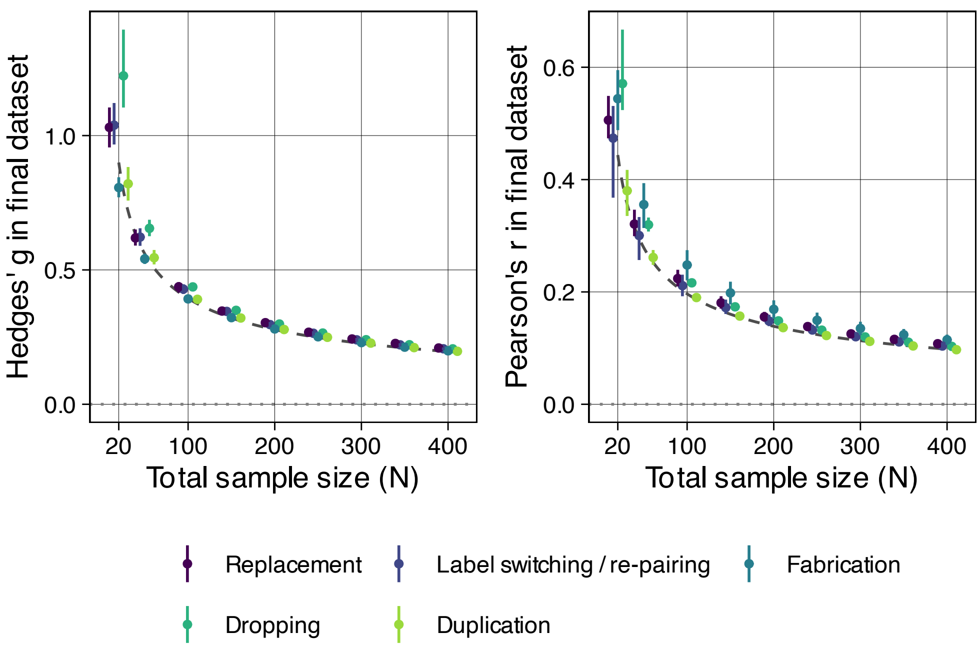

Figure 4: Effect-size bias of the final tampered datasets, relative to the true effect of zero: Hedges' g in the mean-difference setting (left) and Pearson's r in the correlation setting (right). Points are medians across iterations and bars are interquartile ranges; strategies are coloured by family, so that condition switching (mean difference) and re-pairing (correlation) — the two distribution-preserving strategies — share a colour. The dashed curve is the critical effect size — the smallest effect significant at $\alpha = .05$ (two-sided) for the original sample size; strategies that change N (dropping, duplication, fabrication) make this an approximation. Because the greedy procedure stops at the first crossing, the manufactured effects sit just above this curve. Interquartile ranges are shown rather than 95% prediction intervals because the effect-size distribution is a mixture whose small negative component sits almost exactly at the 2.5th percentile (see the text).

That small negative component sits almost exactly at the 2.5th percentile, so a 95% prediction interval intersects it erratically: its lower bound flips sign between adjacent sample sizes according to whether the negative fraction happens to fall just above or just below .025. Figure 4 therefore shows interquartile ranges, which lie well inside the tampered mass and are not subject to that knife-edge.

3. Discussion

3.1. Summary of findings against the aims

The distance from a null dataset to a statistically significant one is short and cheap. The median effort is a small percentage of the sample across strategies and settings, and for most strategies a minority of datasets require more than 10% of the sample to be tampered. Where tampering fails outright it is because it is structurally impossible rather than merely expensive — the rare-outcome 2×2 cells, where too few hostile cases exist to move the table at all. The proportion of the sample that must be altered does not grow with sample size, so the intuition that large studies are safe from this kind of manipulation is wrong in proportional terms; this parallels the finding for *p*-hacking that larger samples alone do not reduce its success rate (Stefan & Schönbrodt, 2023). The strategies that preserve univariate distributions — condition switching and re-pairing, precisely the ones invisible to univariate forensic checks — are among the cheapest. And the effects manufactured by a stop-at-first-crossing procedure are calibrated by construction to look unremarkable.

3.2. Ethics and disclosure

This article documents weaponisable knowledge, and the framing matters: the simulations are designed to make a continuum visible, not to provide a recipe. The manipulations simulated here have been publicly described for years (Brown, 2020; Chambers, 2017); what is added here is quantification.

Consider how the strategies fall under the federal definitions of research misconduct (Steneck, 2007). *Fabrication* is “making up data or results and recording or reporting them” — the fabrication, duplication and replacement strategies simulated here, which add data points that were never collected. *Falsification* is “manipulating research materials, equipment, or processes, or changing or omitting data or results such that the research is not accurately represented in the research record” — the dropping strategy, which omits genuine observations, and the condition switching and re-pairing strategies, which change labels and pairings without adding or deleting anything. The taxonomy that carries the regulatory and moral weight thus classifies by *mechanism*: did you invent it, or did you alter or omit it?

These results say that mechanism is close to irrelevant to what actually matters. Adding invented cases, deleting inconvenient ones, and relabelling existing ones all require a comparable number of edits and leave behind a comparably biased effect. The one clause of the definitions that unifies them is a consequence criterion rather than a mechanism one — *such that the research is not accurately represented in the research record* — and every strategy here satisfies it identically. Each produces a result that did not exist before.

The practical asymmetry, then, is not in the acts but in how they are regarded. Excluding inconvenient participants is admitted by 38–43% of researchers and rated as close to defensible; falsifying data is admitted by under 2% and rated as indefensible (John et al., 2012). When respondents are given provable deniability, admitted fabrication and falsification converge to indistinguishable rates (Gopalakrishna et al., 2022) — evidence that the gap is substantially one of reporting rather than of behaviour. And since perceived seriousness is the strongest available predictor of engagement (Entradas et al., 2026), a seriousness ranking that is mis-set on the dimension these results measure is not merely an intellectual inconsistency: it is a live cause of the behaviour. Deleting a hostile case and inventing a favourable one are, in effort and in damage to the evidence, the same act.

3.3. Limitations

- Proof of principle, not a fraud model. The strategies are deliberately simple and greedy. Real, detection-aware fraud would trade efficiency for plausibility; conversely, several naive strategies here leave detectable signatures (truncated distributions after dropping; leverage outliers after fabrication; shifted marginal prevalence in the 2×2). The distribution-preserving strategies are the existence proof that effective *and* invisible tampering is cheap.
- No detectability axis. I measure effort and bias, not the probability that each manipulation would be caught by modern forensic tools (Brown & Heathers, 2017; Heathers et al., 2018; Wilkinson et al., 2025). Crossing effort with detectability is the most valuable extension.
- A degenerate case in correlation duplication at very small N. With $n = 20$ or 50 the duplication pool is only two or five points, and duplicating them repeatedly can drive the sample to a significant *negative* correlation, which the two-sided test accepts. These iterations are counted as having reached significance even though the manufactured effect points the wrong way. They are a small fraction of successful tampering overall and are confined to the two smallest samples, but a stopping rule requiring the effect to be in the intended direction would exclude them and would be the better specification.
- Greedy myopia is part of the procedure. One-step greedy occasionally stalls — most notably when the initial dataset shows a strong effect opposite to the tampering direction, where a two-sided test's $p \approx 1$ valley cannot be crossed one step at a time. These rare failures are recorded as never attaining significance and are a true property of naive tampering, not an artefact; a less myopic tamperer would do better, so the effort estimates here are, if anything, conservative upper bounds.
- Elementary designs and Normal/binary DGPs. Likert-type outcomes, non-Normal marginals, covariate-adjusted models, and an "alter existing values" strategy distinct from replacement are out of scope. The strategies here also act on the dataset itself; they are therefore complementary to, and not comparable with, the analytic-flexibility strategies compiled by Stefan & Schönbrodt (2023).
- Single threshold. All results are conditioned on $\alpha = .05$, two-sided. No tampering budget is imposed in the design, so results under any budget are recoverable from the recorded counts.

Data and code availability

All simulation code, the full per-strategy research compendia (with extended ADEMP specifications, per-condition tables, and Monte Carlo standard errors for every estimate), and the code that renders this manuscript — every number, figure, and table in it is computed from the simulation results at render time — are available at https://github.com/ianhussey/simulation-impact-of-data-tampering.

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